

Dataset Analysis for Decision-Making

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sicor_municipal_co... .csv
16.7 MB

Analyze the provided dataset to identify its key characteristics for data-driven decision-making. Upon receiving a specific analytical question, address it strictly using the dataset. Your deliverables must include a clear, concise answer to the question and the Python code that extracts and processes the data used to derive your answer.

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To analyze the dataset and provide key characteristics for data-driven decision-making, I'll first outline some general observations about the structure and contents of the dataset. Then, I'll address a specific analytical question using Python code to extract and process the relevant data.

General Observations:

- Structure :** The dataset appears to be tabular with rows representing individual records (e.g., contracts or transactions) and columns representing various attributes associated with each record.
- Columns :** There are numerous columns, but no headers are provided in the raw data. Based on the values, we can infer that:
 - The first column might represent an ID or identifier for the record.
 - Columns include numerical values which could represent monetary amounts, quantities, dates, or other metrics.
 - Some columns contain zero values consistently, suggesting they may be placeholders or unused fields.
 - Several large numeric values suggest financial data like contract amounts or transaction values.
- Data Types :**
 - Most columns seem to contain numerical data (integers or floats).
 - No string/text data is visible, implying categorical data might be encoded numerically.
- Potential Use Cases :**
 - Financial analysis: examining distributions of monetary values, identifying outliers.
 - Operational analysis: tracking quantities or frequencies of certain activities.
 - Time-series analysis: if any columns represent dates/times, trends over time could be explored.

Specific Analytical Question:

Question : What is the total sum of the second-to-last column across all records?

This question seeks to understand the aggregate value of a particular metric, which might represent a key financial or operational figure in the dataset.

Python Code:

Below is the Python code to load the dataset and compute the total sum of the second-to-last column.

```
python
1 import pandas as pd
2
3 # Load the dataset
4 # Assuming the file 'sicor_municipal_contracts.csv' is in the current working directory
5 file_path = 'sicor_municipal_contracts.csv'
6
7 # Load the CSV without headers, as none are provided in the raw data
8 df = pd.read_csv(file_path, header=None)
9
10 # Calculate the total sum of the second-to-last column
11 second_to_last_column_index = df.shape[1] - 2 # Index of the second-to-last column
12 total_sum = df.iloc[:, second_to_last_column_index].sum()
13
```



```
14 # Output the result
15 print(f"The total sum of the second-to-last column is: {total_sum}")
```

Explanation of the Code:

1. **Loading the Data** : The `pandas` library is used to load the dataset into a DataFrame. Since no headers are provided, `header=None` ensures the data is treated as raw numerical entries.
2. **Identifying the Column** : The second-to-last column is identified by calculating its index (`df.shape[1] - 2`), where `df.shape[1]` gives the total number of columns.
3. **Summing Values** : The `.sum()` method computes the total sum of the specified column.
4. **Output** : The result is printed in a clear format.

Deliverable:

After running the code, the output will provide the total sum of the second-to-last column. This value can then be interpreted based on the context of the dataset (e.g., total expenditures, revenues, etc.).

If you have a different specific question or need further analysis, feel free to ask!

How is investment distributed among financial institutions?



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To analyze how investment is distributed among financial institutions in the provided dataset, we need to identify columns that represent:

1. **Financial Institutions** : A column or field indicating the financial institution associated with each record.
2. **Investment Amounts** : A column representing the monetary value of investments or contracts.

Observations from the Dataset:

- The dataset lacks explicit headers, but based on patterns in the data:
 - Column 8 (zero-based index 7) contains large numeric values that could represent investment amounts or contract values.
 - Column 6 (zero-based index 5) appears to contain codes that might correspond to financial institutions or categories.

To determine the distribution of investments among financial institutions, we will:

1. Group the data by the column representing financial institutions.
2. Sum the investment amounts for each group.
3. Analyze and present the results.

Here is the Python code to perform this analysis:

```
python
1 import pandas as pd
2
3 # Load the dataset
4 file_path = 'sikor_municipal_contracts.csv'
5 df = pd.read_csv(file_path, header=None)
6
7 # Define relevant columns
8 institution_column = 5 # Assuming column 6 (zero-based index 5) represents financial institutions
9 investment_column = 7 # Assuming column 8 (zero-based index 7) represents investment amounts
10
11 # Group by financial institution and sum the investment amounts
12 investment_distribution = df.groupby(institution_column)[investment_column].sum()
13
14 # Sort the results in descending order
15 investment_distribution = investment_distribution.sort_values(ascending=False)
16
17 # Display the results
18 print("Investment Distribution Among Financial Institutions:")
19 print(investment_distribution)
```

Explanation of the Code:

1. **Loading the Data** : The dataset is loaded into a Pandas DataFrame without headers (`header=None`).



- 2. **Defining Columns** : We assume column 6 (index 5) represents financial institutions and column 8 (index 7) represents investment amounts.
- 3. **Grouping and Summing** : Using `groupby` , we aggregate the investment amounts for each unique value in the financial institution column.
- 4. **Sorting** : The results are sorted in descending order to show the institutions with the highest investments first.
- 5. **Output** : The distribution is printed, showing the total investment for each financial institution.

Example Output:

The output will look something like this (values are illustrative):

```
1 Investment Distribution Among Financial Institutions:
2 502      50000000.0
3 201      45000000.0
4 430      30000000.0
5 300      25000000.0
6 505      20000000.0
7 ...
8 Name: 7, dtype: float64
```

Interpretation:

- The output shows the total investment amount for each financial institution.
- For example, if `502` has the highest value, it indicates that institution `502` received the largest share of investments.

Next Steps:

If the dataset includes additional context (e.g., names of institutions or categories), we can map the codes to meaningful labels for better interpretability. If you have specific requirements or additional details about the dataset, please provide them for further refinement of the analysis.

